Hooke/history normal impact

LAMMPS dem01 | Two equal spheres | 29 October 2026

PASS	All four runs pass official 2% restitution and momentum checks.

Official rebound checks at 83.85 us

Run	Measured e	Error (%)	Left vx (m/s)	Right vx (m/s)	Official
CPU dt	0.4996646896	0.067062	-0.999329379	0.999329379	PASS
GPU dt	0.4996646896	0.067062	-0.999329379	0.999329379	PASS
CPU dt/2	0.4998413098	0.031738	-0.999682620	0.999682620	PASS
GPU dt/2	0.4998413098	0.031738	-0.999682620	0.999682620	PASS

Target e = 0.5; official relative tolerance = 2%. The momentum bound is abs(m v_left + m v_right) <= 0.02 (2m) vin = 1.413716694e-05 kg m/s. Both checks apply at step 2795. Pinned saved-state e = 0.499664689634 is a separate reference.

Official inputs and direct mapping

Quantity	Unchanged input / FunDEM configuration
Two free spheres	Each R = 0.0025 m; density = 2700 kg/m^3; centers x = +/-0.00255 m; vx = +/-2 m/s toward contact; all other motion and initial spin zero.
Hooke/history law	kn = 1e6 N/m; kt = 2.8e5 N/m; e = 0.5; sliding friction mu = 0. Explicit linear stiffness enabled.
Prescribed damping	dampingEnabled = true: explicit cn = 4.0504686872 N s/m; ct = 0 and both rotational damping coefficients = 0. Normal cn is derived from official e without fitting.
Other settings	No gravity, wall, global damping or constraints; rolling/torsional resistance = 0. E = 1e4 Pa and nu = 0.33 are inactive linear-stiffness schema placeholders.
Time / full output	dt = 30 / 15 ns; 2795 / 5590 steps; same 83.85 us endpoint; output every integration step including t = 0.

Normal coefficient and tangential scope

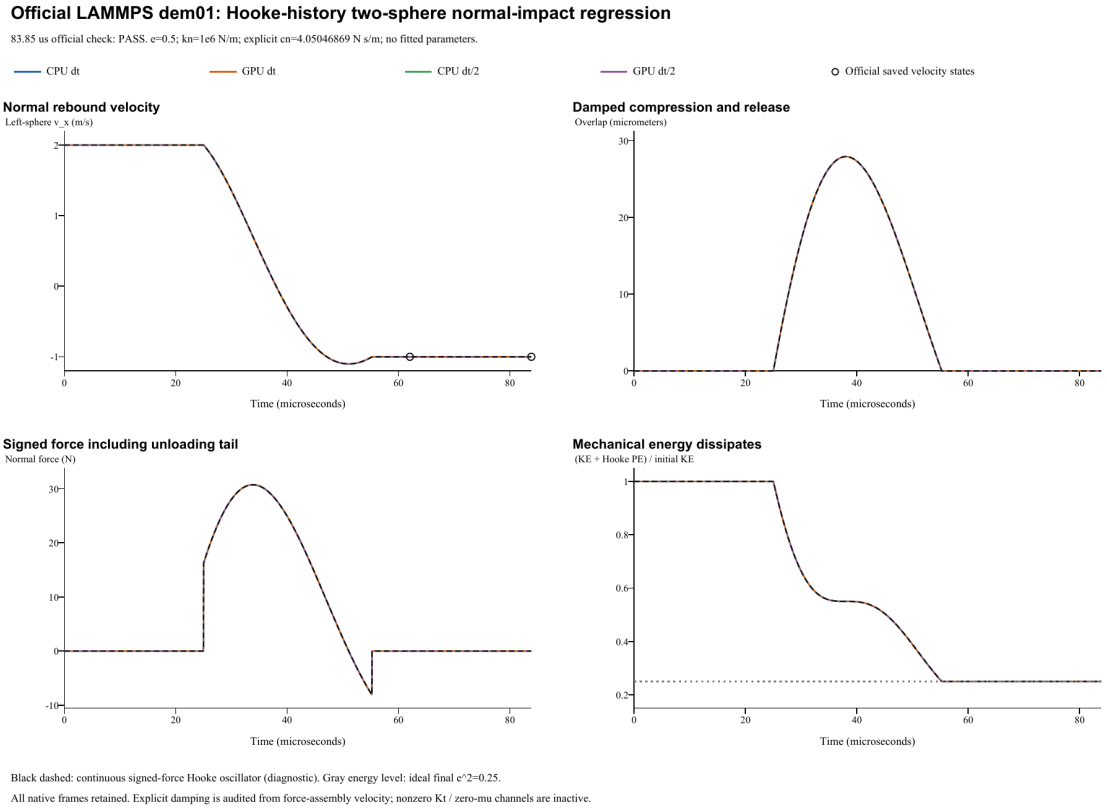
m* = m/2 = 8.8357293382e-05 kg; beta = -ln(e)/sqrt(pi^2 + ln(e)^2).
cn = 2 beta sqrt(m* kn) = 4.0504686872 N s/m; Fn = kn delta - cn vn_assembly.
Signed unloading force is retained; no limit_damping clipping or solver changes.

The pair-level manual switch preserves official ct = 0 with nonzero kt. Material e = 0.5 remains recorded, but this run audits the explicit coefficient and adds no automatic-damping-branch verification. With zero slip, zero spin and mu = 0, tangential history remains inactive; active friction/history evolution is outside scope.

Reference provenance

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 6900a6efd5cf80f862da2241c2b48e7d4cbaf7f1. LAMMPS was not rerun locally. YAML epsilon 1e-10 is same-code regression precision, not a universal cross-code tolerance.

Native histories and retained evidence



Prescribed damping and two-body response

Run	Inferred cn (N s/m)	Min. Fn (N)	Final KE / initial	Max. momentum
CPU dt	4.050468687	-8.104974	0.249664802	0
GPU dt	4.050468687	-8.104974	0.249664802	0
CPU dt/2	4.050468687	-8.092331	0.249841335	0
GPU dt/2	4.050468687	-8.092331	0.249841335	0

Force-based inference agrees with prescribed cn to a maximum relative error of 7.44e-15; maximum force-reconstruction residual is 3.55e-15 N. The coefficient is an inferred diagnostic, not a direct native output channel.

CPU/GPU whole-history maximum velocity / signed-normal-force differences are 4.44e-16 m/s / 7.11e-15 N. Halving dt changes CPU restitution by 0.00017662.

Maximum pair momentum-impulse balance residual is 2.93e-18 kg m/s. Final KE is compared with e^2 times initial KE = 0.25; dissipative contact is not expected to conserve mechanical energy.

Maximum tangential force / sliding-spring component / spin are 0 N / 0 m / 0 rad/s. Newly executed trajectories differ from corresponding prior Hooke no-history runs by at most 4.44e-16 m/s in velocity; this is a zero-slip normal-response diagnostic.

Inference removes the last half-kick: vn_assembly = vn_saved - dt Fn/(2m*). Samples use abs(vn_assembly) > 0.08 m/s and delta > 1e-10 m. Black dashed oscillator curves are supplemental. Zero tangential force/history and spin are checked; the nonzero kt is inactive in this normal collision.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

2796 frames per base run; 5591 per half-step run: all 16,774 native frames, contacts, input projects, every-step CSVs, audits and backend logs are retained. All four runs were newly executed.

Run manifest | Analysis JSON | Checkpoint CSV | Full native ZIP

Execution and scope

Real SimulationSession runs use linked FunDEM 0.4.3 CPU/CUDA native libraries; installed GUI executable byte identity was not established. Scope: damped normal Hooke impact, explicit restitution-derived damping, signed unloading force, momentum and inactive tangential history. Active tangent/history, spin and many-particle flows require separate cases.